

Problem 2. Set Formulas and Propositional Formulas

(a) Use a truth table to verify that the propositional formula $(P \wedge \overline{Q}) \vee (P \wedge Q)$ is equivalent to P .

P	Q	$\overline{Q}$	$P \wedge Q$	$P \wedge \overline{Q}$	$(P \wedge Q) \vee (P \wedge \overline{Q})$
T	T	F	T	F	T
F	T	F	F	F	F
T	F	T	F	T	T
F	F	T	F	F	F

As shown, the first column P and the last column $(P \wedge Q) \vee (P \wedge \overline{Q})$ are exactly the same. Therefore, they are equivalent.

(b) Prove that $A = (A - B) \cup (A \cap B)$ for all sets A, B , by showing $x \in A$ IFF $x \in ((A - B) \cup (A \cap B))$.

Proof. We will prove this by showing both directions of the implication.

Forward implication ($\implies$): We want to show:

$$x \in A \implies x \in ((A - B) \cup (A \cap B))$$

Assume $x \in A$. We consider two possible cases for B :

- **Case 1:** $x \notin B$. Since $x \in A$, by the definition of set difference, $x \in (A - B)$. By the definition of union, if an element is in a set, it must also be in the union of that set with any other set. Therefore, $x \in ((A - B) \cup (A \cap B))$.
- **Case 2:** $x \in B$. Since $x \in A$ and $x \in B$, by the definition of intersection, $x \in (A \cap B)$. By the definition of union, it follows that $x \in ((A - B) \cup (A \cap B))$.

In both cases, the implication holds.

Backward implication ($\impliedby$): We want to show:

$$x \in ((A - B) \cup (A \cap B)) \implies x \in A$$

By the definition of union, we can say $x \in (A - B)$ or $x \in (A \cap B)$.

- **Case I:** $x \in (A - B)$. By the definition of set difference, if an element exists in the set difference $(A - B)$, then that element must be part of the first set (A). Thus, $x \in A$.
- **Case II:** $x \in (A \cap B)$. By the definition of set intersection, $x \in A$ and $x \in B$. This immediately implies $x \in A$.

Since $x \in A$ holds in both cases, this implication also holds.

Both the backward and forward implications are true. Subsequently, we have proven that $x \in A$ IFF $x \in ((A - B) \cup (A \cap B))$. $\square$

Relation to part (a):

It is clear that part (a) and part (b) mirror each other. Part (b) is the set theory equivalent of the propositional logic in part (a).

If we directly compare them:

- $(A - B)$ is the equivalent of $(P \wedge \overline{Q})$.
- $(A \cap B)$ is the equivalent of $(P \wedge Q)$.
- $\cup$ (union) is the equivalent of $\vee$ (OR).

Problem 3. Predicate Practice

(a) The predicate $\text{sixish}(n)$ defined as “ n is a natural number whose first digit is 6.”

$$\text{sixish}(n) := \exists k. (6 \times 10^k \leq n < 7 \times 10^k)$$

(b) “There are exactly two natural numbers n such that $Q(n)$ is true.”

$$\exists a, b. (a \neq b \wedge Q(a) \wedge Q(b) \wedge \forall c. (a = c \vee b = c))$$

(c) Goldbach’s Conjecture: “every even integer greater than 2 can be written as the sum of two primes.”

$$\forall n. (n > 2 \wedge \text{isEven}(n) \implies \exists p, q. (\text{isPrime}(p) \wedge \text{isPrime}(q) \wedge n = p + q))$$

(d) “Every prime larger than 100 is sixish.”

$$\forall n. (n > 100 \wedge \text{isPrime}(n) \implies \text{sixish}(n))$$

(e) $\text{No}, \text{R}(n)$ is not equivalent to $\text{isOdd}(n)$ in domain of $\mathbb{N}$

proof by counterexample. Try $n = 3$: $\text{isOdd}(3)$ is true since $3 = 2(1) + 1$

but in case of $\text{R}(n)$, $3 = 2b + 5 \implies b = -1$ which is not possible since $b \in \mathbb{N}$ Thus proving $\text{isOdd}(n)$ and $\text{R}(n)$ are not equivalent $\square$

Problem 4. Induction and Contradiction

(a)

$P(n) :=$ for every integer $n \geq 1$ and any n numbers $x_1, x_2, \dots, x_n \in \mathbb{R}, Q(x_1 \cdot x_2 \cdot \dots \cdot x_n) \implies Q(x_1) \vee Q(x_2) \vee \dots \vee Q(x_n)$

Given: For all $x, y \in \mathbb{R}$,

$$Q(x \cdot y) \implies (Q(x) \vee Q(y)). \quad (1)$$

This property will be used during Inductive step.

proof by Induction. :We prove $P(n)$ by doing induction on n .

Base case ($n = 1$): There is only one value so, $Q(x_1) \implies Q(x_1)$ is always true.

Inductive step: Assume $P(n)$ is true for an arbitrary $n \in \mathbb{N}(n \geq 1)$, now we want to prove $P(n + 1)$ i.e

$$Q(x_1 x_2 \dots x_n x_{n+1}) \implies \bigvee_{i=1}^{n+1} Q(x_i)$$

Where $x_{n+1} \in \mathbb{R}$. Let $y = (x_1 x_2 \dots x_n)$, then

$$Q(y \cdot x_{n+1}) \implies Q(y) \vee Q(x_{n+1})$$

Now from given property (1) we can say that implication holds and from induction hypothesis we know $Q(y) \implies \bigvee_{i=1}^n Q(x_i)$ and by substituting it we get,

$$Q(y \cdot x_{n+1}) \implies \left(\bigvee_{i=1}^n Q(x_i) \right) \vee Q(x_{n+1})$$

also by expanding the value of y and simplifying the RHS of implication,

$$Q(x_1 x_2 \dots x_n x_{n+1}) \implies \bigvee_{i=1}^{n+1} Q(x_i)$$

Which is equivalent to $P(n + 1)$, so $P(n + 1)$ must be true, and by principle of mathematical induction we conclude that $P(n)$ is true for every $n \geq 1$. $\square$

(b),(c) The definition of $Q(x)$ that we should use is:

$$Q(x) :- \nexists p, q \in \mathbb{Z}, q \neq 0 : x = \frac{p}{q}$$

OR

$$Q(x) :- x \text{ is irrational.}$$

Lamma 1: If product of two real number is irrational then atleast one of the two numbers must be irrational.

proof by contradiction: Assume $\exists x, y \in \mathbb{R}$ $Q(x \cdot y)$ is true and x and y both are rational. but that immediately creates a contradiction because as we know product of two rational is always rational which contradicts our assumption.

□

Theorem 1. If the product of $n \geq 1$ real numbers is irrational, then at least one of those numbers must be irrational.

Proof. From lamma 1 we have already proven that $Q(x \cdot y) \implies Q(x) \vee Q(y)$ and by following the proof in (a) we can conclude that if the product of $n \geq 1$ real numbers is irrational, then at least one of those number must be irrational.

□